School of Computer Science and Artificial Intelligence

Lab Assignment # 6.5

Program : B. Tech (CSE)

Specialization :AIML

Course Title : AI Assisted Coding

Course Code : 23CS002PC304

Semester : VI

Academic Session : 2025-2026

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Batch No. : 33

Experiment 6: AI-Based Code Completion

Aim

To use AI-based code completion tools to generate, analyze, optimize, and ethically evaluate Python programs involving classes, loops, and conditional statements.

Learning Outcomes Addressed

- LO1: Generate Python code using AI
- LO2: Explain AI-generated code line-by-line
- LO3: Identify logical flaws or inefficiencies
- LO4: Optimize AI-generated code
- LO5: Demonstrate ethical use of AI tools

Task 1: Conditional Eligibility Check (Voting Eligibility)

Prompt:

"Generate Python code to check voting eligibility based on age and citizenship."

Code:

```
age = int(input("Enter your age: "))
citizen = input("Are you a citizen? (yes/no): ").lower()

if age >= 18 and citizen == "yes":
    print("You are eligible to vote.")
else:
    print("You are not eligible to vote.")
```

Output:

```
Enter your age: 20
Are you a citizen? (yes/no): yes
You are eligible to vote.
```

Explanation:

- Takes age as integer input

- Takes citizenship status as string

- Uses **AND condition**:

  - Age must be **18 or above**
  - Citizenship must be **yes**

- Prints eligibility result

<u>Task 2:</u> Loop-Based String Processing (Vowels & Consonants)

"Generate Python code to count vowels and consonants in a string using a loop."

Code:

```
text = input("Enter a string: ").lower()

vowels = "aeiou"

vowel_count = 0

consonant_count = 0


for char in text:

    if char.isalpha():

        if char in vowels:
```

```
            vowel_count += 1
        else:
            consonant_count += 1

print("Vowels:", vowel_count)
print("Consonants:", consonant_count)
```

Output:

```
Enter a string: sowmya
Vowels: 2
Consonants: 4
```

Explanation:

- Converts string to lowercase
- Loops through each character
- Checks:
    - Alphabet or not
    - Vowel or consonant
- Counts correctly

<u>Task 3:</u> AI-Generated Library Management System

Prompt:

> *"Generate a Python program for a library management system using classes, loops, and conditional statements."*

Code:

```
class Library:
```

```
    def __init__(self):
        self.books = []

    def add_book(self, book):
        self.books.append(book)
        print(f"'{book}' added to library.")

    def display_books(self):
        if not self.books:
            print("No books available.")
        else:
            print("Books in library:")
            for book in self.books:
                print("-", book)


library = Library()

while True:
    print("\n1. Add Book")
    print("2. Display Books")
    print("3. Exit")

    choice = input("Enter choice: ").lower().strip()

    if choice in ["1", "add book", "add"]:
```

```
        book = input("Enter book name: ")
        library.add_book(book)

    elif choice in ["2", "display books", "display"]:
        library.display_books()

    elif choice in ["3", "exit"]:
        print("Exiting program.")
        break

    else:
        print("Invalid choice. Please try again.")
```

Output:

```
1. Add Book
2. Display Books
3. Exit
Enter choice: 1
Enter book name: Harry Potter
'Harry Potter' added to library.

1. Add Book
2. Display Books
3. Exit
Enter choice: 1
Enter book name: Hunger Games
'Hunger Games' added to library.

1. Add Book
2. Display Books
3. Exit
Enter choice: 2
Books in library:
- Harry Potter
- Hunger Games

1. Add Book
2. Display Books
3. Exit
```

```
1. Add Book
2. Display Books
3. Exit
Enter choice: 3
Exiting program.
```

Explaination:

AI greatly speeds up initial code creation and helps structure programs logically. However, human review is essential to verify correctness, improve features, and ensure ethical use without blindly trusting generated code.

<u>Task 4:</u> Class-Based Attendance System

Prompt:

> *"Generate a Python class to mark and display student attendance using loops."*

Code:

```
class Attendance:
    def __init__(self):
        self.students = {}

    def mark_attendance(self, name, status):
        self.students[name] = status

    def display_attendance(self):
        for name, status in self.students.items():
```

```
        print(name, ":", status)

attendance = Attendance()

attendance.mark_attendance("Alice", "Present")
attendance.mark_attendance("Bob", "Absent")
attendance.display_attendance()
```

Output:

```
Alice : Present
Bob : Absent
```

Explanation:

-Dictionary stores student attendance

- Loop prints all records

- Class improves data organization

<u>Task 5:</u> ATM Menu Simulation (Loops & Conditionals)

Prompt:

"Generate a Python program using loops and conditionals to simulate an ATM menu."

Code:

```
balance = 5000

while True:
```

```
print("\nATM Menu")
print("1. Check Balance")
print("2. Deposit")
print("3. Withdraw")
print("4. Exit")

choice = input("Enter choice: ")

if choice == "1":
    print("Balance:", balance)

elif choice == "2":
    amount = int(input("Enter deposit amount: "))
    balance += amount
    print("Amount deposited.")

elif choice == "3":
    amount = int(input("Enter withdrawal amount: "))
    if amount <= balance:
        balance -= amount
        print("Withdrawal successful.")
    else:
        print("Insufficient balance.")

elif choice == "4":
    print("Thank you!")
```

```
        break

    else:
        print("Invalid option.")
```

Output:

```
ATM Menu
1. Check Balance
2. Deposit
3. Withdraw
4. Exit
Enter choice: 2
Enter deposit amount: 20000
Amount deposited.

ATM Menu
1. Check Balance
2. Deposit
3. Withdraw
4. Exit
Enter choice: 1
Balance: 25000

ATM Menu
1. Check Balance
2. Deposit
3. Withdraw
4. Exit
Enter choice: 3
Enter withdrawal amount: 10000
Withdrawal successful.
```

Explanation:

- AI was used as a **coding assistant**, not a replacement
- Code was **reviewed, tested, and optimized manually**
- No blind copying—logic was understood and explained
- Encourages learning, not dependency